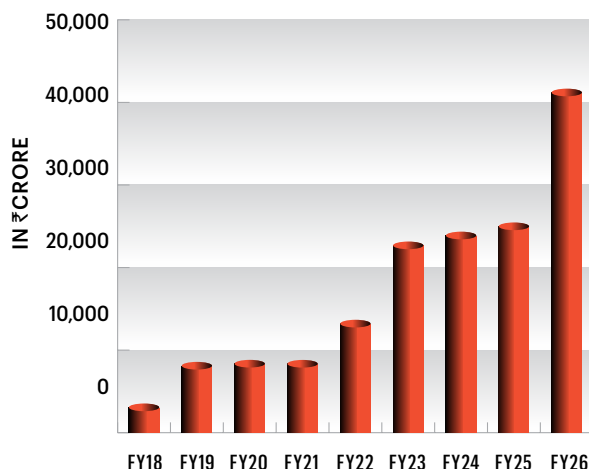


THE CELL SURGE

● India's lithium-ion cell imports up 12x since FY18



SOURCE COMMERCE MINISTRY

THE RARE EARTH PERMANENT MAGNETS SCHEME (REPMS)

Capacity

Creation: 6,000 MTPA integrated manufacturing capacity of sintered REPMs, distributed among up to five beneficiaries through global competitive bidding

Capital Subsidy:

₹750 crore to support advanced facilities

Timeline: Two-year gestation for setting up, followed by five years of incentive disbursement linked to production

Objective:

Establish an end-to-end ecosystem from rare-earth oxides to finished magnets, ensuring supply for sectors like electric mobility, renewable energy, electronics, aerospace, and defence

₹7,280

CRORE OUTLAY FOR THE SCHEME

₹6,450

CRORE SALES-LINKED INCENTIVES, OVER FIVE YEARS

Gujarat's Sanand is expected to start production in 2027.

NO CUSTOMERS

Developing a domestic ecosystem for upstream materials such as cathode and anode is turning out to be a chicken-and-egg situation as raw material producers await cell manufacturing to start.

"My customers are struggling because they don't have their own technology and are getting bits and pieces. Increasingly, Chinese are restricting partnerships for technology. They don't want to give anything. Across anode, cathode and cell, the Chinese government is not allowing partnerships with Indian companies," says Vikram Handa, founder and managing director of Epsilon Advanced Materials.

Manohar says the greatest chal-

lenge arises when moving from promising laboratory results to reliable industrial-scale production during which safety, consistency, degradation control and cycle life are the key to success. "The toughest manufacturing challenge in battery cell production is achieving consistent yield at scale. Due to the extremely precision-intensive process, every step must be tightly controlled. Small deviations can quickly translate into defects, safety concerns and lower yields," says Manohar.

For the Indian climate, the ideal chemistry is LFP because of lower overheating risk. "The challenge remains in technology. LFP today is only concentrated in China," says Handa. LFP also reduces the country's dependence on nickel and cobalt.

"Technology know-how is lack-

ing. You can buy equipment from Japan and Korea with longer lead time. From China, it was available off the shelf. It's a solvable problem. What's not a solvable problem is how to run the factory, how to make a good cell," says Handa.

India's EV ambitions are vulnerable, says Handa. "Graphite is one thing that is totally China-centric. Tomorrow, if they say no graphite for the rest of the world, it's not like I can pay \$1 more and buy it from other countries. It's just not available. Its processing technology and capacity don't exist. If I was putting ₹4,000-5,000 crore into building a gigafactory in India, I would ensure multiple sources for the material that is most concentrated in China. I can buy lithium from five different places, only the price varies. This is something for which I don't have sources," he says.